

$$f(x) = \frac{1}{\Gamma(r)} e^{-x} x^{r-1} dx, \quad x > 0$$

3_ The Gamma Distribution: $X \sim \text{Gamma}(\lambda, r)$

$$f(t) = \frac{\lambda^r t^{r-1}}{\Gamma(r)} e^{-\lambda t}, \quad t > 0, \lambda > 0, r > 0$$

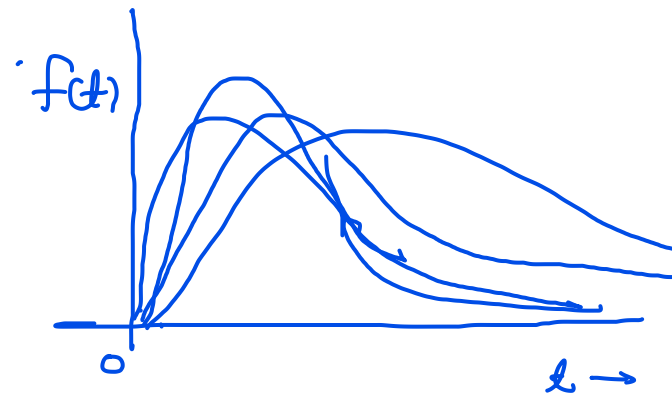
$$\int_{-\infty}^{\infty} f(t) dt = \int_0^{\infty} \frac{\lambda^r t^{r-1}}{\Gamma(r)} e^{-\lambda t} dt$$

$$r=1 \\ f(t) = \lambda e^{-\lambda t}, \quad t > 0$$

$$= \frac{1}{\Gamma(r)} \int_0^{\infty} (\lambda t)^{r-1} e^{-\lambda t} d(\lambda t)$$

$$= \frac{1}{\Gamma(r)} \int_0^{\infty} y^{r-1} e^{-y} dy$$

$$= 1.$$



$$E(T^k) = \int_0^{\infty} t^k \cdot \frac{\lambda^r t^{r-1}}{\Gamma(r)} e^{-\lambda t} dt$$

$$= \frac{\lambda^r}{\Gamma(r)} \int_0^{\infty} t^{k+r-1} e^{-\lambda t} dt = \frac{\lambda^r}{\Gamma(r) \lambda^{k+r}} \int_0^{\infty} (\lambda t)^{k+r-1} e^{-\lambda t} d(\lambda t)$$

$$= \frac{\lambda^r}{\Gamma(r)} \frac{\Gamma(k+r)}{\lambda^{k+r}}$$

$$\text{or, } E(T^k) = \frac{\Gamma(k+r)}{\Gamma(r)} \cdot \frac{1}{\lambda^k}, \quad k = 1, 2, 3, \dots$$

$$E(T) = \frac{\Gamma(r+1)}{\Gamma(r)} \cdot \frac{1}{\lambda} = \frac{r}{\lambda} \quad \checkmark$$

$$E(T^2) = \frac{\Gamma(r+2)}{\Gamma(r)} \cdot \frac{1}{\lambda^2} = \frac{(r+1)r \Gamma(r)}{\Gamma(r)} \cdot \frac{1}{\lambda^2} = \frac{r(r+1)}{\lambda^2}$$

$$\begin{aligned}
 \therefore V(T) &= E(T^2) - (ET)^2 \\
 &= \frac{r(r+1)}{\lambda^2} - \frac{r^2}{\lambda^2} \\
 &= \left(\frac{r}{\lambda^2} \right)
 \end{aligned}$$

$$\begin{aligned}
 M_T(u) &= E(e^{uT}) = \int_0^{\infty} e^{ut} \cdot \frac{\lambda^r}{\Gamma(r)} e^{-\lambda t} t^{r-1} dt \\
 &= \frac{\lambda^r}{\Gamma(r)} \int_0^{\infty} e^{-t(\lambda-u)} \cdot t^{r-1} dt \\
 &= \lambda^r \cdot \frac{1}{(\lambda-u)^r}, \quad u < \lambda \\
 &= \left(\frac{\lambda}{\lambda-u} \right)^r, \quad u < \lambda
 \end{aligned}$$

Ex: The CPU time requirement T , for jobs has a Gamma distribution with mean 40 and standard deviation 20 seconds. Any job taking less than 20 second is a short job. What is the probability that ^{out} of 5 randomly jobs, at least 2 are short jobs?

Soln: Given $\frac{r}{\lambda} = 40$, $\frac{r}{\lambda^2} = 400 \Rightarrow r = 4$
 $\lambda = \frac{1}{10} \checkmark$

$$\underline{P(T < 20)} = \int_0^{20} \frac{1}{\Gamma(4)} e^{-\frac{t}{10}} t^3 dt$$

$$= 1 - P(T \geq 20)$$

$$= 1 - \int_{20}^{\infty} \frac{1}{6 \cdot 10^4} e^{-\frac{t}{10}} t^3 dt$$

$$\frac{t}{10} = y \\ \Rightarrow \frac{1}{10} dt = dy$$

$$= 1 - \int_2^{\infty} \frac{1}{6} e^{-y} y^3 dy$$

$$= 1 - \frac{19}{8} e^{-2} = 0.1429$$

Z : RV denote the no of short jobs

$$Z \sim \text{Bin}(5, 0.1429)$$

$$P(Z=z) = \binom{n}{z} p^z (1-p)^{n-z},$$

$$z=0, 1, 2, \dots, n$$

$$P(Z \geq 2) = 1 - P(Z=0) - P(Z=1)$$

$$= 1 - (1 - 0.1429)^5 - 5(1 - 0.1429)^4(0.1429)$$

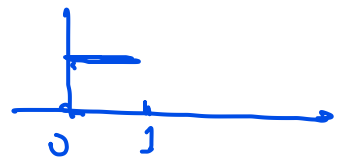
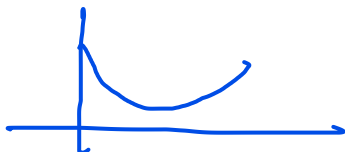
$$= \underline{0.1519}$$

4. Beta distribution:

A continuous RV X is said to have a Beta distribution with parameters α & β if it has pdf given by

$$f_X(x) = \begin{cases} \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}, & 0 < x < 1, \alpha > 0, \beta > 0 \\ 0, & \text{otherwise} \end{cases}$$

$$\alpha=0, \beta=0, f_X(x) = 1, 0 < x < 1$$



$$E(x^k) = \int_0^1 x^k \cdot \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1} dx$$

$$= \frac{1}{B(\alpha, \beta)} \int_0^1 x^{k+\alpha-1} (1-x)^{\beta-1} dx$$

$$= \frac{1}{B(\alpha, \beta)} B(k+\alpha, \beta)$$

$$= \frac{(\alpha+k-1)(\alpha+k-2) \dots (\alpha+1)\alpha}{(\alpha+\beta+k-1)(\alpha+\beta+k-2) \dots (\alpha+\beta)}$$

$$B(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)}$$

$$\Gamma m = (m-1) \Gamma m-1$$

$$E(x) = \frac{\alpha}{\alpha+\beta}$$

$$E(x^2) = \frac{\alpha(\alpha+1)}{(\alpha+\beta)(\alpha+\beta+1)}$$

$$\therefore V(x) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)} \checkmark$$